

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

EE3002-1 – LINEAR CONTROL SYSTEMS

(E&E)

Duration: 3 Hours

Max. Marks: 100

Note:

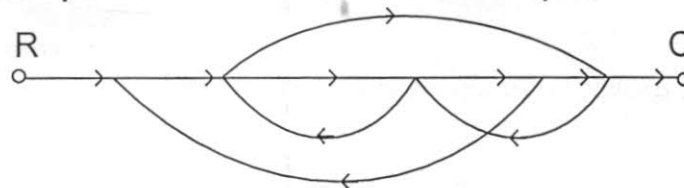
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

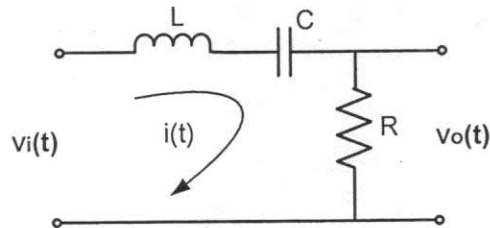
20 Marks

- In open loop system the control action
 - depends on the size of the system
 - depends on system variables
 - depends on the input signal
 - is independent of the output
- An automatic toaster is a _____ loop control system.
 - open
 - closed
 - partially closed
 - any of the above
- In force-voltage analogy, mass is analogous to _____.
 - charge
 - current
 - inductance
 - resistance
- The total number of forward path and number of individual loop for the SFG is _____



- 2,3
 - 2,4
 - 2,5
 - 1,4
- In a system zero initial condition means that
 - The system is at rest and no energy is stored in any of its components
 - The system is working with energy stored in the elements
 - The system is working with zero reference signal
 - None of these
 - Velocity error constant of a system is measured when the input to the system is unit _____ function.
 - parabolic
 - ramp
 - impulse
 - step
 - The position and velocity errors of a type-1 system are _____.
 - constant, constant
 - constant, infinity
 - zero, constant
 - zero, zero
 - The type 0 system has _____ at the origin.
 - no pole
 - net pole
 - simple pole
 - two poles
 - For a unity feedback system with $G(s)=4/(s(s+1))$ the undamped natural frequency is _____.
 - 4
 - 2
 - 6
 - 7
 - The condition of the system represented by the polynomial $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$
 - Stable
 - Marginally stable
 - Unstable
 - Equilibrium state

11. An open loop transfer function with negative unity feedback is given by $G(s) = \frac{K(s+1)}{s(s+2)(s+5)}$. The number of asymptotes will be ____
- A) 3 B) 2
C) 4 D) 1
12. In a second order system the locations of poles for undamped response will be ____
- A) Right half of the s-plane B) Left half of the s-plane
C) On the imaginary axis D) At the origin
13. In root locus plot, the breakaway points ____
- A) must lie outside the root loci B) should always be on the real axis plane
C) must lie on the root loci D) Must be between 0 and -1
14. The transfer function $\frac{V_o(t)}{V_i(t)}$ for the given circuit is ____



Given $R=1\ \Omega$, $L=1\text{H}$ and $C=1\text{F}$

- A) $\frac{V_o(s)}{V_i(s)} = \frac{2s}{s^2 + s + 1}$ B) $\frac{V_o(s)}{V_i(s)} = \frac{s}{s^2 + s + 2}$
C) $\frac{V_o(s)}{V_i(s)} = \frac{s}{s^2 + s + 1}$ D) $\frac{V_o(s)}{V_i(s)} = \frac{s}{s^2 + 2s + 1}$
15. The response of the system if the characteristic equation of the closed loop system is $s^2 + 4s + 25 = 0$
- A) Over damped B) Undamped
C) Critically damped D) Under damped
16. All the asymptotes cross the real axis at a point called
- A) centroid B) Break in point
C) Breakaway point D) Stable point
17. With respect to a lag compensator which of the statements are true.
- (i) large improvement in steady state performance.
(ii) zero is located to the left of the pole on negative real axis.
(iii) Suppress the steady state performance.
(iv) zero is located to the right of the pole on negative real axis
- A) (i) and (ii) B) (iii) only
C) (iii) and (iv) D) (iv) only
18. Frequency at which the magnitude plot crosses the 0 decibel line is ____
- A) phase cross over frequency B) Gain margin
C) Gain cross over frequency D) Phase margin
19. Which among the following constitute the state model of a system in addition to state equations?
- A) Input equations B) State trajectory
C) Output equations D) State vector
20. State space analysis is applicable even if the initial conditions are ____
- A) Zero B) Equal
C) Non-zero D) Not equal

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	With an example illustrate open loop and closed loop control system.	5	L*2	1	1
b)	Obtain the input-output relationship of following mechanical systems:				
i.)	Mass				
ii.)	Spring				
iii.)	Damper	3	L1	1	1

- c) Using Block diagram reduction technique, find the overall transfer function for the system shown in Fig: Q1c.

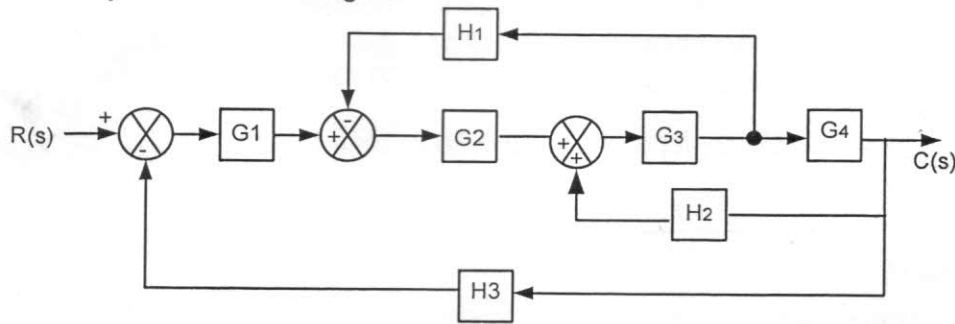


Fig: Q1c

8 L3 2 1

2. a) For the mechanical system shown in Fig: Q2a obtain the force-voltage analogous network.

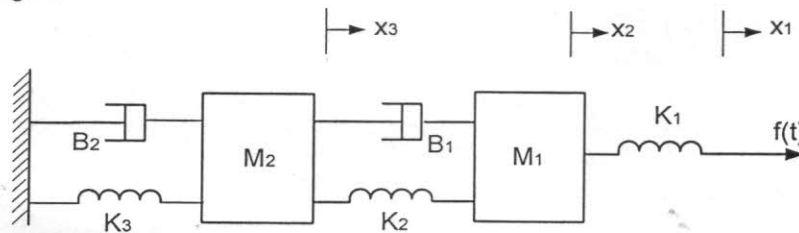


Fig: Q2a

8 L3 1 1

- b) Derive the expression for the transfer function for armature-controlled DC motor and hence draw the block diagram.

8 L3 2 1,2

3. a) The open loop transfer function of a unity feedback system is:

$$G(s) = \frac{4}{s(s+1)}$$

Determine the nature of the response of the closed loop system for a step input. Also determine rise time and peak time

8 L3 1 1,2

- b) Obtain the transfer function for the given signal flow graph in Fig: Q3b using Mason's gain formula.

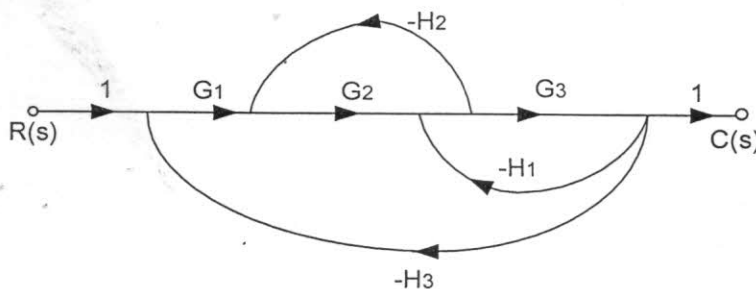


Fig: Q3b

8 L2 2 1

Unit – II

4. a) The characteristic equation for a certain control feedback system is given as $s^4 + 5s^3 + 5s^2 + 4s + k = 0$. Determine the range of k for which the system is stable.

6 L3 3 1

- b) Sketch the Bode plot for the open loop transfer function.

- $G(s) = K$
- $G(s) = Ks$
- $G(s) = \frac{1}{(1+sT)}$
- $G(s) = (1+sT)$

10 L2 4 1

5. a) State and explain Routh stability criterion. 4 L1 3 1
 b) A negative unity feedback control system has open loop transfer function consisting of two poles, two zeros and a variable gain K. The zeros are located at -2 and -1. The poles are located at -0.1 and 1. Using Routh stability criterion determine the range of values of K for which the closed loop system is stable. 6 L4 3 1,2
 c) Determine the centroid and angle of departure for the given loop transfer function.

$$G(s) = \frac{k(s+9)}{(s^2+4s+11)}$$

6 L3 4 1,2

6. a) Draw the root locus of the system whose open loop transfer function is given by

$$G(s) = \frac{K}{s(s+2)(s+4)}$$

10 L3 3 1,2

- b) Write a note on the following terms.
 (i) gain cross over frequency (iii) Phase cross over frequency
 (ii) phase margin (iv) gain margin

06 L1 4 1

Unit – III

7. a) Elaborate on lag compensator and lead compensator with Bode plot. 8 L2 5 1
 b) With neat diagram explain PI, PD and PID controllers. 8 L2 5 1
 8. a) Write a note on following.
 (i) State (ii) State variable (iii) State vector (iv) state space 8 L1 5 1
 b) For the electrical circuit shown in Fig Q8b, obtain state space model.

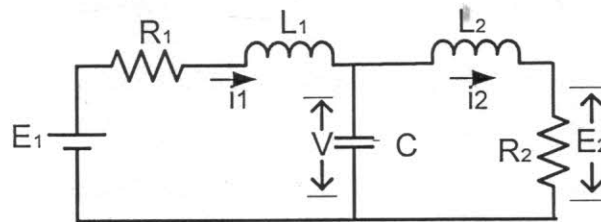


Fig: Q8b

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
